

Practice problems #6

1. Find the B -matrix for the transformation $x \mapsto Ax$, when $B = \{b_1, b_2\}$.

$$A = \begin{bmatrix} 3 & 4 \\ -1 & -1 \end{bmatrix}, \quad b_1 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad b_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

2. Consider the matrix

$$A = \begin{bmatrix} 4 & 3 & 1 & 1 \\ 3 & 4 & 1 & 1 \\ 1 & 1 & 4 & 3 \\ 1 & 1 & 3 & 4 \end{bmatrix}.$$

Suppose $T(x) = Ax$. Is there a basis B such that $[T]_B$ is diagonal? If so, find the B -matrix for T . If not, say why.

Answers or Hints on page 2.

#1: If $T(x) = A(x)$, what's $T(b_1)$?

Ans:

$$\begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix}$$

#2: There can be many different answers. One possible answer:

$$[T]_B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 9 \end{bmatrix}.$$